

A LAGRANGIAN BETA DISTRIBUTION

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SUMMARY

A new generalization known as the Lagrangian Beta (LB) distribution is developed in the sense given by Nelson and Consul (1975) for the Lagrangian gamma distribution. It is the distribution of the time between occurrences of a generalized negative binomial (GNB) process as described by Jain and Consul (1971). Moments of the LB distribution are derived. This distribution has four parameters n , r , α and β and tables and graphs for the density and distribution functions are given for $n = 2; 3; 4$; $r = 2; 4$; $\alpha = 0, 1(0, 1)1, 0$ and $\beta = 1(1)5$.

1. Introduction

The present paper develops a generalization of a beta distribution on the basis of the discrete Lagrange distributions. One of these, the generalized negative binomial (GNB) distribution recently studied in detail by Jain and Consul (1971) is given by

$$(1.1) \quad p(x; n, \alpha) = \frac{n}{x!} \frac{\Gamma(n + \beta x)}{\Gamma(n + \beta x - x + 1)} \cdot \alpha^x (1 - \alpha)^{n + \beta x - x} \quad \text{for } x = 0; 1; \dots$$

where $n > 0$, $0 < \alpha < 1$, $|\alpha\beta| < 1$. By assigning suitable values to the parameters, this GNB distribution can be used to describe data that follow binomial as well as Poisson distributions.

In the next section we develop a continuous distribution for the waiting time between the occurrences of a GNB process. Since the time between the occurrences of a classical negative binomial process follows a beta distribution, we call this new distribution the Lagrangian Beta (LB) distribution. It has been shown that the classical beta distribution is a particular case of this LB distribution for integer r when $\beta = 1$. Furthermore, it tends to a classical exponential distribution for any value of β when r is unity as $n \rightarrow \infty$. Some other properties of the LB distribution are discussed. Also some graphical representations of the LB density are made.

2. Derivation of the LB distribution

It is known that the time between occurrences of a classical negative binomial pro-

cess follows a beta distribution. Now, let us suppose that the process is given by the GNB (1.1). Then change α to αt in (1.1) where $t \geq 0$ is the time between occurrences of the GNB process, $0 \leq \alpha t \leq 1$. With this notation, the cumulative distribution of the random variable T , which is the time until the r -th occurrence of the GNB process, is given by

$$\begin{aligned}
 F(t; n, \alpha, \beta, r) &= P(T \leq t) \\
 (2.1) \quad &= \sum_{x=r}^{\infty} \frac{n}{n+\beta x} \binom{n+\beta x}{x} (\alpha t)^x (1-\alpha t)^{n+\beta x-x}, \quad 0 \leq \alpha t \leq 1 \\
 &= 1 - \sum_{x=0}^{r-1} \frac{n}{n+\beta x} \binom{n+\beta x}{x} (\alpha t)^x (1-\alpha t)^{n+\beta x-x}
 \end{aligned}$$

This will be called the Lagrangian beta (LB) distribution.

Note that $\sum_{x=0}^{r-1} \frac{n}{n+\beta x} \binom{n+\beta x}{x} (\alpha t)^x (1-\alpha t)^{n+\beta x-x}$ is a decreasing function of t for given α and β . Then $f(1/\alpha; n, \alpha, \beta, r) = 1$.

The density function of T , obtained by differentiating (2.1) with respect to t , can be written as

$$\begin{aligned}
 (2.2) \quad f(t; n, \alpha, \beta, r) &= \sum_{x=0}^{r-1} \frac{n}{n+\beta x} \binom{n+\beta x}{x} \alpha (\alpha t)^{x-1} (1-\alpha t)^{n+\beta x-x-1} \\
 &\quad \cdot [\alpha t(n+\beta x) - x], \quad 0 \leq \alpha t \leq 1
 \end{aligned}$$

which is a linear sum of classical beta distributions and may thus also be termed a generalized beta distribution.

Three cases arise from this:

- (i) For $r = 1$, $n \rightarrow \infty$ and for any value of β , (2.2) reduces to the classical exponential distribution with density function:

$$(2.3) \quad f(t; n, \alpha, \beta, 1) = K e^{-Kt}, \text{ where } K = n\alpha.$$

- (ii) When $\beta = 1$, (2.2) reduces to

$$(2.4) \quad f(t; n, \alpha, 1, r) = \frac{\alpha}{B(n, r)} (\alpha t)^{r-1} (1-\alpha t)^{n-1},$$

which is a classical beta distribution for any positive integer r . It may thus be expected that many natural phenomena may be better described by the LB distribution than by the usual beta distribution.

- (iii) Letting $\alpha = a/n$ and $\beta = n\theta$, where a and θ are fixed, the LB distribution reduces asymptotically (as $n \rightarrow \infty$) to the form

$$(2.5) F(t; a, \theta, r) = 1 - \sum_{x=0}^{r-1} \frac{at[(1+x\theta)at]^{x-1}}{x!} \cdot e^{-(1+x\theta)at},$$

which is called a Lagrangian gamma (LG) distribution by Nelson and Consul (1975).

3. Estimation of parameters

The estimation of the parameters involved in the LB distribution can be made with the help of the moment expressions given below.

The j -th moment of the underlying distribution is given by

$$(3.1) E(T^j) = \sum_{x=0}^{r-1} \frac{n}{n+\beta x} \binom{n+\beta x}{x} \alpha \left[(n+\beta x) \int_0^{1/\alpha} t^j (\alpha t)^x (1-\alpha t)^{n+\beta x-x-1} dt \right. \\ \left. - x\alpha \int_0^{1/\alpha} t^j (\alpha t)^{x-1} (1-\alpha t)^{n+\beta x-x-1} dt \right],$$

which becomes, after further simplification,

$$E(T^j) = \frac{n}{\alpha^j} \left[\sum_{x=0}^{r-1} \frac{(x+j)(x+j-1) \dots (x+1)}{(n+\beta x-x)(n+\beta x+j) \dots (n+\beta x+1)} \right. \\ \left. - \sum_{x=1}^{r-1} \frac{(x+j-1)(x+j-2) \dots (x)}{(n+\beta x-x)(n+\beta x+j-1) \dots (n+\beta x)} \right]$$

In particular, the first and the second moments are given by

$$(3.2) E(T) = \frac{n}{\alpha} \sum_{x=0}^{r-1} \frac{1}{(n+\beta x)(n+\beta x+1)},$$

which is the mean time to the occurrence of a GNB process, and

$$(3.3) E(T^2) = \frac{2}{n\alpha} E(T) + \frac{2}{\alpha^2} \sum_{x=0}^{r-1} \frac{x(n-\beta) - 2}{(n+\beta x)(n+\beta x+1)(n+\beta x+2)}$$

We note from (3.2) that the mean of the LB distribution decreases as β increases and vice versa.

These expressions become more difficult to solve when the parameter r is large. As an alternative way of estimating the parameters by the method of moments, one can use the inverse moments, namely, $E(T^{-1})$, $E(T^{-2})$, etc., given by

$$(3.4) E(T^{-1}) = n\alpha \left[\frac{(n+\beta+r-2)}{(r-1)(n+\beta+1)} + \sum_{j=0}^{\infty} \frac{\Gamma(n+\beta)}{\Gamma(\beta-j-1) \Gamma(n+j+1)} \frac{(-1)^j}{(j+1)} B(n, \beta) \right],$$

for $r \geq 2$;

and

$$(3.5) E(T^{-2}) = n\alpha^2 \left[\frac{(n+2\beta)(n+2\beta-1)}{2(n+2\beta-2)} - 2 \sum_{x=3}^{r-1} \frac{(n+\beta x-1)}{x(x-1)(x-2)} \right] \\ + n\alpha^3 \cdot \frac{1}{\alpha} \sum_{j=0}^{\infty} \left\{ \binom{\beta-1}{j+2} + (\beta-1) \binom{\beta-1}{j+1} \right\} (-1)^j (\alpha t)^j (1-\alpha t)^{n-1} dt$$

for $r \geq 3$.

In a similar way, other higher inverse moments can be obtained. The expressions for inverse moments are sufficient to provide an alternative means of estimating the parameters involved in the LB distribution when the expectations are replaced by the corresponding sample moments.

4. Graphical Representations

The LB density values and distribution function values were obtained by computing

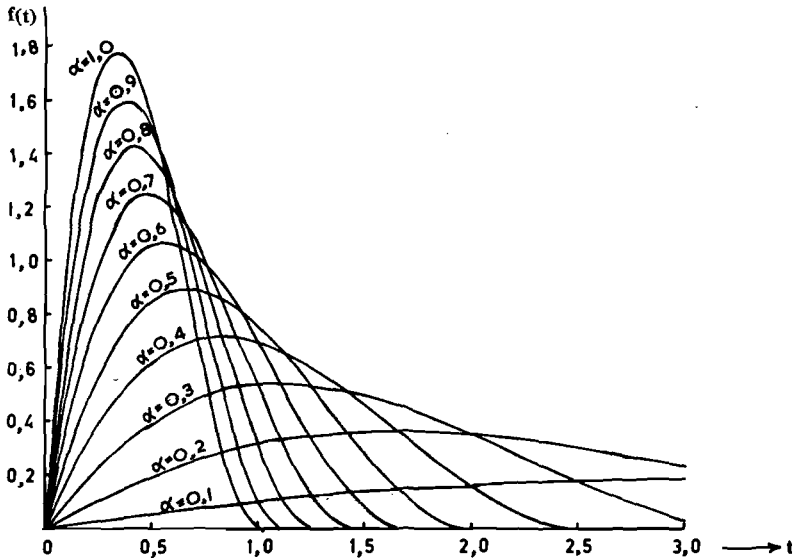


Fig. 1: LB density for $N=3$; $\beta=1$; $r=3$ and $\alpha=0,1(0,1)1,0$

tables for the following set of parameter values:

$$\begin{aligned}
 n &= 2; 3; 4; \\
 r &= 2; 4; \\
 \alpha &= 0,1(0,1)1,0; \\
 \beta &= 1; 2; 3; 4; 5; \\
 \text{and } t &= 0(0,5)10,0 \quad \text{for } \alpha = 0,1 \\
 &= 0(0,3)4,8 \quad \text{for } \alpha = 0,2 \\
 &= 0(0,2)3,2 \quad \text{for } \alpha = 0,3 \\
 &= 0; 0,1; 0,2; \dots \quad \text{for } \alpha = 0,4(0,1)1,0
 \end{aligned}$$

such that $0 \leq \alpha t \leq 1$.

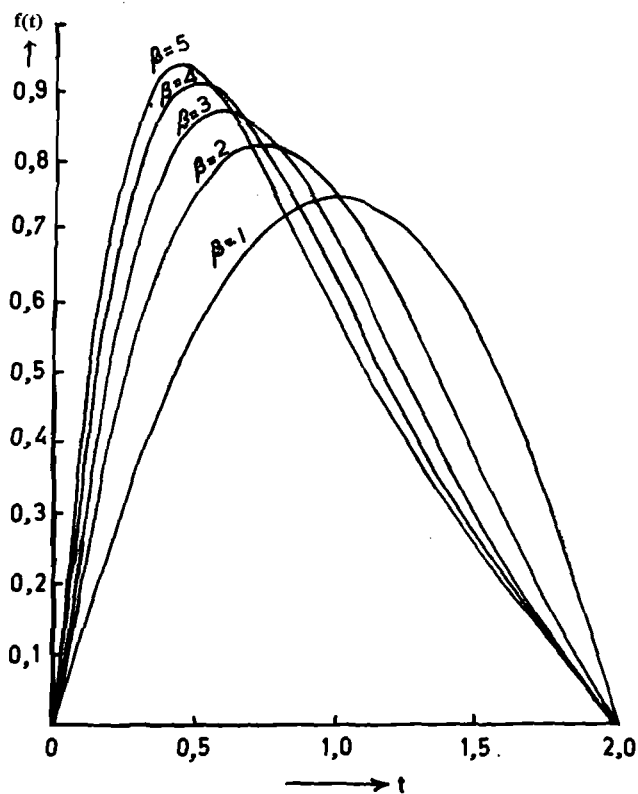


Fig. 2: LB density for $N=2$; $\alpha=0,5$; $r=2$ and $\beta=1; 2; 3; 4; 5$

To prepare the tables for density values and distribution values a FORTRAN computer programme was developed (which may be obtained upon request from the authors). With the help of these tables the behaviour of the LB distribution was investigated by graphical representations. Figures 1; 2; 3 and 4 show the graphs drawn from these tables for typical values of the parameters.

Figure 1 shows the densities for fixed n and β and for several values of α . It may be seen that as α increases, the mean, mode, median, and variance all decrease and the density becomes increasingly peaked. The top of the density curves become more and more a narrow and peaked one. The positive skewness of the curve also diminishes rapidly. Furthermore, for fixed n and several values of β and α , the shape of the curve becomes more peaked as β and α increase.

Figure 2 indicates the behaviour of the densities for fixed n and α and for several values of β . As stated above, the curve becomes slowly peaked as β increases. At the same time the mean, mode and variance also decrease. It is symmetric when $\beta = 1$ and thus it is a classical beta distribution.

Figure 3 demonstrates how the density takes its form for fixed values of α and β , but for several values of n . We note here that the curves are positively skew and such skewness becomes less as n increases. Moreover, the peakedness of the curve increases as n increases. Thus the fact that the mean of the distribution as well as its mode decrea-

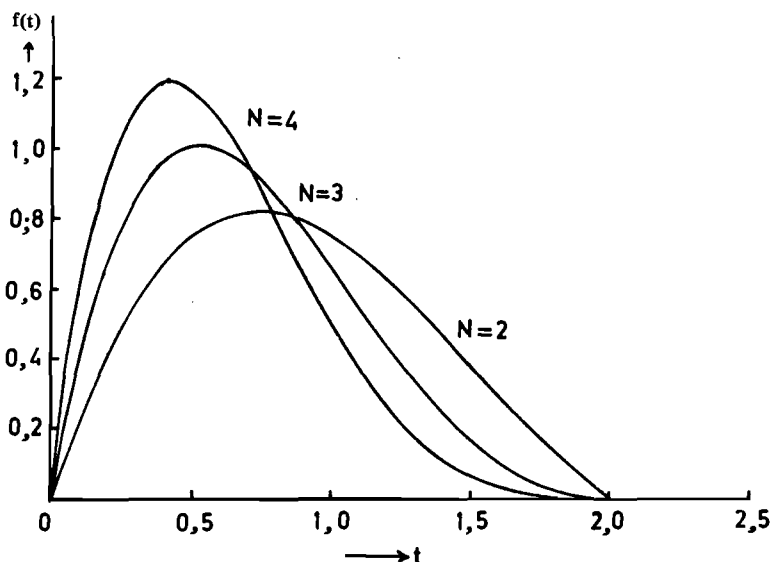


Fig. 3: LB density for $\beta=2$; $\alpha=0.5$; $r=2$ and $N=2,3,4$

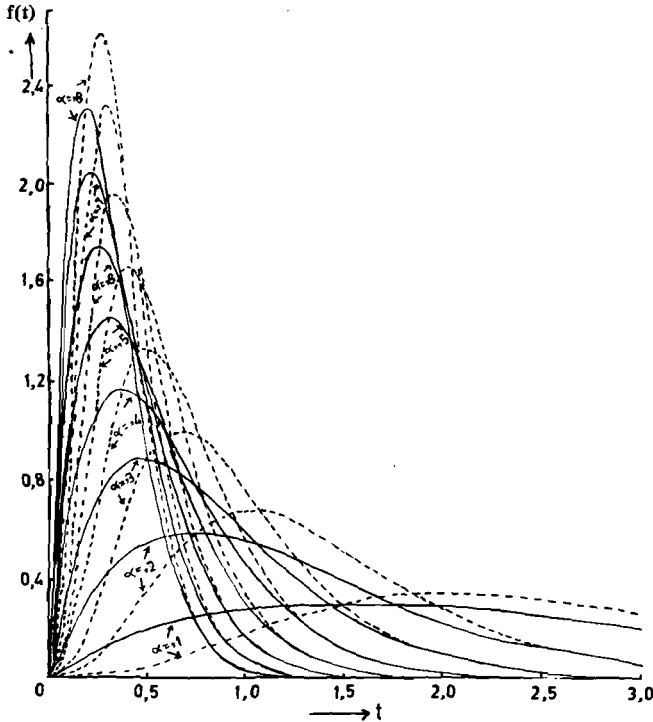


Fig. 4: LB density for $N=4$; $\beta=5$; $\alpha=0,1(0,1)0,8$ when $r=2$ and $r=4$

ses as n increases is clearly recognizable.

Figure 4 provides some kind of comparison for fixed values of n , β and α and for varying values of r . It is observed that the flatness of the curve decreases and it becomes more peaked as r increases.

From all these typical graphical representations it is evident that the shape of the density curve becomes more peaked and its skewness diminishes rapidly as the parametric values increase. And thereby it seems the shapes of the curves are tending to resemble the well-known standard normal distribution.

5. Other comments

If we use the transformation $z = \alpha t$ in equation (2.2), it gives a standard form of the LB distribution in three parameters only:

$$(5.1) \quad f(z; n, \beta, r) = \sum_{x=0}^{r-1} \frac{n}{n+\beta x} \binom{n+\beta x}{x} z^x (1-z)^{n+\beta x-x} \cdot [z(n+\beta x) - x]$$

and corresponding values of $E(X)$, $E(X^{-1})$, $E(X^2)$, $E(X^{-2})$, etc., can be derived from the expressions already given earlier, by setting $\alpha = 1$. Thus α is merely a scale parameter in the LB distribution similar to σ in the normal distribution. This standard form of the LB density can be used for the tabulation of probability points.

The LB distribution considered here should find a variety of applications in waiting line models and queueing theory, where the arrival process can be described by a generalized negative binomial process (including binomial process). The fitting and other properties of this distribution will be published in a separate paper.

REFERENCES

- CONSUL, P.C. and JAIN, G. C. (1973). A generalization of the Poisson distribution, *Technometrics*, **15**, 791-799.
- JAIN, G. C. and CONSUL, P.C. (1971). A generalized negative Binomial distribution, *SIAM J. Appl. Math.*, **21**, 501-513.
- NELSON, D.L. and CONSUL, P.C. (1975). A Langrangian Gamma Distribution, *Statistical distributions in scientific work, Vol. I - Models and Structures*, Patil, Kotz and Ord (eds.), D. Reidel, Dordrecht and Boston, 241-246.

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